Oresajo Oluwaferanmi Idunnuoluwa

Mobile Software Engineer · Flutter · Kotlin · Swift

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Google Play (my apps) | TestFlow (live product)

SUMMARY

Flutter developer on TestFlow, a CBT exam platform with **5,000+ users (1,500+ active)**: I work across the Flutter app and the Node.js API, including FCM push in 500-token multicast batches and a MongoDB-backed shared rate limiter. Built a **GPS-verified clock-in** for Forge (Flutter, Google Maps, Squad wallet flows) in a 4-person hackathon team. One app live on Google Play (ClassBook). Software Engineering student at OAU.

SKILLS

Mobile	Flutter, Dart, Riverpod, Kotlin, Jetpack Compose, Swift, Room, WorkManager, Google Maps SDK, Geolocator (GPS), NFC, BLE, Firebase Cloud Messaging, local notifications
Backend	Node.js, Express, REST APIs, JWT auth, CSRF, rate limiting, Zod validation, Go, Python
Data	MongoDB, PostgreSQL, Redis, Firebase, Supabase
Payments	Paystack, Payaza, Squad (wallet and withdrawal flows in Forge)

EXPERIENCE

Software Engineer, TestFlow — Present · Lagos, Nigeria

Computer-based testing (CBT) platform for university students. 5,000+ users, 1,500+ active. Flutter app, React PWA, Node.js API on MongoDB.

- Engineer across the Flutter mobile app (Riverpod, Dio) and a Node.js/Express REST API of **300+ routes backed by 250+ automated tests**, shipping timed exams, auto-saved answers, instant results, review and performance analytics for 5,000+ students.
- Built push notifications on **Firebase Cloud Messaging** using the Firebase Admin SDK: device tokens are registered from the app and stored by the backend, and messages are sent in multicast batches of 500 tokens so large broadcasts stay inside provider rate limits.
- Built **real-time live exam rooms** over WebSockets with a Server-Sent Events fallback: one isolated room per exam, event fan-out to participants, and clients that reconnect after a server restart and re-sync state over REST so no exam state is lost.
- Integrated **Paystack** payments for student passes and organisation subscriptions: bank-transfer checkout, webhooks verified by HMAC on the raw request body, server-side re-verification and amount and currency checks before any access is granted, idempotent checkout, and prorated upgrades and renewals.
- Designed abuse protection for authentication and password reset: a **MongoDB-backed shared rate limiter** that holds across every API instance, per-IP limits on auth, upload and checkout routes, and hashed device-fingerprint and IP signals to stop repeat claims from throwaway accounts and rotating browsers.
- Secured the API with JWT sessions, CSRF protection, Helmet, Zod request validation, bcrypt password hashing and an audit log, covered by unit and integration tests, and designed to stay correct across multiple API instances.

SELECTED PROJECTS

Forge – GPS job marketplace for blue-collar workers (Flutter mobile app) — GitHub

Team hackathon project (4 contributors). I built the Flutter worker app; the NestJS/PostgreSQL API and web dashboards were built by teammates.

- Built **GPS-verified clock-in**: the worker must have location on and be at the job site. The app sends live coordinates to the backend, which confirms they are within the site and the assigned shift window before work time starts, which cuts off-site clock-ins and pay claims made while away.
- Built nearby-job discovery on Google Maps with live device location (`geolocator`, `google_maps_flutter`), plus the wallet, withdrawals and loan screens that sit on the Squad payment flow.
- Integrated Firebase Cloud Messaging push notifications; app state managed with Riverpod.

Flutter · Dart · Riverpod · Geolocator · Google Maps · FCM

MyDrop – native file sharing between Mac and Android GitHub

- Built it because an Android phone could not send files to a Mac in any clean way. Native **Swift** macOS app and **Kotlin / Jetpack Compose** Android app that speak the Quick Share protocol over Wi-Fi and work with compatible devices.
- Added a Finder Share extension, notifications, folder zip-and-send, clipboard sync, notification mirroring, trusted devices, QR sharing for people without the app, and phone files browsable in Finder. Used it to learn Apple's Liquid Glass design system. Released separately for macOS and Android (v1.0.0).

Swift · Kotlin · Jetpack Compose · Quick Share · macOS Share extensions

TapPay – NFC, QR and identity-based payment wallet 1st place, hackathon at Redeemer's University

- Flutter wallet that adapts to whatever the situation allows: **NFC tap**, QR codes or transfer by identity, with a record of every transaction. Integrated **Payaza** for payments.
- Won first place at the 2026 hackathon at Redeemer's University, Ede.

Flutter · Dart · NFC · Payaza

ClassBook – study library for course reps and classmates Live on Google Play

- Published Flutter app where course reps and classmates share PDFs and materials, and students keep course PDFs in one organised, searchable library to read, highlight and sync with their class. Shipped to the Play Store and maintained, with separate APK builds for direct install and testing.

Flutter · Dart · Google Play

Hydro – heat, UV and air-quality companion for Nigeria (Android) GitHub

- Native Android app that turns raw weather data into decisions: when to leave, how much water to carry, which hours are risky. Risk numbers are computed by deterministic code from published formulas (NOAA/NWS heat index, WHO UV index); the Gemini model only chooses tools and explains results, and stale data is flagged instead of shown as a false all-clear.
- Map-first interface with live location, route-aware advice, offline cache, and quiet-hours alerts with de-duplication.

Kotlin · Jetpack Compose · Room · WorkManager · Google Maps Platform · Gemini API

fhare – Android phone as a real second display for a Mac GitHub

- Not screen mirroring: macOS registers a genuine virtual display, captured on the GPU with ScreenCaptureKit, hardware-encoded with VideoToolbox (H.265/H.264), streamed over Wi-Fi and decoded with MediaCodec in a Flutter app, with touch and typing sent back to the Mac.

Swift · Flutter · ScreenCaptureKit · VideoToolbox · MediaCodec

Tara – AI participant for live meetings (Python backend) GitHub

- Voice agent that joins a meeting and takes part only as far as the organiser's rules and the live state of the conversation allow. Deterministic code owns the floor, permissions, identity and tools; the model only proposes. Handles interruptions, speaker tracking, meeting minutes and document generation.

Python · Gemini Live · PostgreSQL

Mailx – open-source email infrastructure (Go) GitHub

- Go service for sending mail through a tenant-scoped REST API or authenticated SMTP, with direct MX or relay delivery, DKIM signing, recipient suppression, signed webhooks and durable retry history. PostgreSQL for delivery records, Redis for scheduling, Docker for local setup, GitHub Actions for CI and security checks.

Go · PostgreSQL · Redis · SMTP · Docker

EDUCATION

B.Sc. Software Engineering, Obafemi Awolowo University, Ile-Ife, Nigeria In progress

AWARDS & CERTIFICATIONS

- **First place**, 2026 hackathon at Redeemer's University, Ede, with TapPay.
- Flutter & Dart Development Bootcamp (Dr. Angela Yu, Udemy), 2025.